Project Title

Live Currency Converter with Trend Chart (Python)

1. Problem Statement

People often need to convert money between currencies, but checking rates online every time is slow. This project builds a simple Python program that converts amounts using **live exchange rates** and shows a **trend chart** for the selected currency pair.

2. Objectives

- Fetch live exchange rates for 170+ currencies from a free API.
- Convert any amount from one currency to another.
- Provide a user-friendly command-line interface.
- Display a simple trend chart for the selected pair.

3. Technologies Used

- **Python 3.x**
- **requests** – to fetch live rates from https://api.exchangerate-api.com.
- **matplotlib.pyplot** – to draw the trend chart.

4. How It Works

1. The user enters a **base currency** (like USD), **target currency** (like INR), and an **amount**.
2. The program fetches the latest exchange rate from the API and computes:

$$\text{converted} = \text{amount} \times \text{rate}$$

3. It prints the result and shows a **trend chart** for the last 7 days (using sample data; can be replaced with real historical data later).
4. The loop continues until the user types quit.

5. Key Features

- Uses **real-time exchange rates** from the web.
- Works with **170+ currencies**.
- Handles **errors** like invalid input or network issues.
- Shows a **trend chart** and saves it as currency_trend.png.

6. Limitations

- The **trend uses sample data** instead of real history.
- **Command-line only**, no GUI.
- Depends on a **free API** that may have limits.

7. Conclusion

This project combines **API use, basic math, and visualization** into a small but useful currency converter. The code is clean and easy to extend—for example, by adding a GUI, real historical data, or a conversion history system.